ACCOUNTS & FINANCE EXAMINATION PRACTICE SERIES • JKSSB FAA & SSC CGL AAO

Cost Accounting: 50 High-Yield Exam MCQs

Detailed Question Bank with Formulas, Practical Numericals & Conceptual Explanations

CORE COST SHEET ARCHITECTURE MATRIX

Cost Component	Constituent Elements / Formula	Exam Treatment / Trap
Prime Cost	Direct Material + Direct Labour + Direct Expenses	Directly traceable variable manufacturing costs.
Works / Factory Cost	Prime Cost + Factory Overheads + Opening WIP - Closing WIP	Work-in-progress (WIP) adjusted strictly at factory level.
Cost of Production	Factory Cost + Administration / Office Overheads	Basis for valuation of unsold finished goods stock.
Cost of Sales (Total Cost)	Cost of Goods Sold (COGS) + Selling & Distribution Overheads	Exclude Income Tax, Debenture Interest, Dividends.

SECTION 1: COST CLASSIFICATION & BEHAVIORAL CONCEPTS (Q1 – Q7)

Q1. A past expenditure that has already occurred, cannot be altered by any present or future management decision, and is irrelevant for decision-making is termed:

(a) Opportunity Cost
(b) Sunk Cost
(c) Marginal Cost
(d) Imputed Cost

Answer: (b) Rationale: Sunk cost represents historical, unrecoverable outlays (e.g., past market survey fees or historical book cost of discarded machinery) that must be ignored when evaluating future operational choices.

Q2. The maximum potential benefit lost or forgone by selecting one operational alternative over the next best available alternative is defined as:

(a) Replacement Cost
(b) Opportunity Cost
(c) Out-of-Pocket Cost
(d) Standard Cost

Answer: (b) Rationale: Opportunity cost is the measurable sacrifice of the next best forgone alternative. For instance, using owned factory premises rather than leasing them out entails an opportunity cost equal to the lost rental income.

Q3. When the total production volume increases within the relevant operating range, what is the behavior of fixed cost per unit and variable cost per unit?

(a) Fixed cost/unit increases; Variable cost/unit decreases
(b) Fixed cost/unit decreases; Variable cost/unit remains constant
(c) Both fixed and variable cost per unit remain constant
(d) Both fixed and variable cost per unit decrease

Answer: (b) Rationale: Total fixed cost is constant, so fixed cost per unit declines as volume rises ($FC / Units$). Variable cost varies in total proportionally with activity, maintaining a constant rate per unit.

Q4. A telephone bill containing a fixed monthly line rental of ₹400 plus an additional usage charge of ₹1.20 per call beyond 100 free calls is classified as a:

(a) Purely Fixed Cost
(b) Purely Variable Cost
(c) Semi-Variable (Mixed) Cost
(d) Discretionary Fixed Cost

Answer: (c) Rationale: Semi-variable (or mixed) costs possess both a fixed component (incurred regardless of activity) and a variable component (fluctuating with output/usage).

Q5. Hypothetical or notional costs that do not involve any physical cash disbursement but are recognized for managerial economic decision-making are called:

(a) Imputed (Notional) Costs
(b) Explicit Costs
(c) Committed Costs
(d) Conversion Costs

Answer: (a) Rationale: Imputed costs (e.g., interest on owner's proprietary capital, salary of an unpaid entrepreneur) involve no actual cash outlay but reflect true economic resource usage.

Q6. In cost accounting terminology, 'Conversion Cost' is defined mathematically as:

(a) Direct Material + Direct Labour
(b) Direct Labour + Direct Expenses + Factory (Works) Overheads
(c) Factory Cost + Office Administration Overheads
(d) Prime Cost + Selling Overheads

Answer: (b) Rationale: Conversion cost is the total expenditure required to transform raw material into finished product: $Direct\ Labour + Direct\ Expenses + Factory\ Overheads$ (i.e., Production Cost excluding Direct Material).

Q7. The aggregate sum of all indirect costs (Indirect Material + Indirect Labour + Indirect Expenses) is technically known as:

(a) Prime Cost
(b) Overheads (Oncost)
(c) Opportunity Cost
(d) Sunk Cost

SECTION 2: ELEMENTS OF COST & COST SHEET ARCHITECTURE (Q8 – Q15)

Q8. Given: Opening Stock of Raw Material = ₹40,000, Purchases = ₹1,80,000, Carriage Inwards = ₹10,000, and Closing Stock = ₹30,000. Raw Material Consumed equals:

(a) ₹2,00,000
(b) ₹1,90,000
(c) ₹2,10,000
(d) ₹1,80,000

Answer: (a) Rationale: *Raw Material Consumed = Opening Stock (₹40,000) + Purchases (₹1,80,000) + Carriage Inward (₹10,000) - Closing Stock (₹30,000) = ₹2,00,000*. Carriage inward is an essential direct procurement cost.

Q9. Which of the following financial items is STRICTLY EXCLUDED from preparation of a Cost Sheet?

(a) Carriage on purchase of raw materials
(b) Factory power and fuel
(c) Dividend paid to equity shareholders
(d) Showroom sales commission

Answer: (c) Rationale: Purely financial appropriations (dividends paid, corporate income tax, interest on debentures, loss on sale of capital assets, goodwill write-offs) are excluded from cost accounts.

Q10. In a Cost Sheet, Work-in-Progress (WIP) adjustments (Opening WIP added, Closing WIP deducted) are made at which stage?

(a) To arrive at Prime Cost
(b) To arrive at Factory (Works) Cost
(c) To arrive at Cost of Goods Sold
(d) To arrive at Cost of Sales

Answer: (b) Rationale: Work-in-Progress occurs on the factory floor; hence, *Gross Factory Cost + Opening WIP - Closing WIP = Net Factory / Works Cost*.

Q11. Prime Cost equals ₹3,00,000; Factory Overheads are 20% on Prime Cost; Office Overheads are 10% on Factory Cost. The Cost of Production is:

(a) ₹3,96,000
(b) ₹3,60,000
(c) ₹4,10,000
(d) ₹4,20,000

Answer: (a) Rationale: Factory Overheads = $20\% \times ₹3,00,000 = ₹60,000$. Factory Cost = $₹3,00,000 + ₹60,000 = ₹3,60,000$. Office Overheads = $10\% \times ₹3,60,000 = ₹36,000$. *Cost of Production* = $₹3,60,000 + ₹36,000 = ₹3,96,000$.

Q12. Lubricating oil, cotton waste, and cleaning materials utilized on factory machinery are classified as:

(a) Direct Material
(b) Indirect Material
(c) Direct Expenses
(d) Administrative Overheads

Answer: (b) Rationale: Consumable stores such as lubricants and rags do not form a physical part of the finished saleable product; hence, they constitute indirect material (factory overhead).

Q13. Remuneration paid to an expert for a specialized engineering patent design prepared exclusively for a specific client's custom job is treated as:

(a) Direct Expenses (Chargeable Expenses)
(b) Factory Overhead
(c) Administrative Overhead
(d) Selling Overhead

Answer: (a) Rationale: Direct or chargeable expenses are those outlays (other than direct material and direct labor) that are incurred specifically and directly for a particular job, forming part of Prime Cost.

Q14. If Cost of Production is ₹5,00,000, Opening Finished Goods = ₹50,000, and Closing Finished Goods = ₹70,000, what is the Cost of Goods Sold (COGS)?

(a) ₹5,20,000
(b) ₹4,80,000
(c) ₹5,50,000
(d) ₹4,70,000

Answer: (b) Rationale: *COGS = Cost of Production (₹5,00,000) + Opening Finished Goods (₹50,000) - Closing Finished Goods (₹70,000) = ₹4,80,000*.

Q15. Total Cost of Sales is ₹8,00,000 and profit is desired at 20% on Selling Price. The total Selling Price should be:

(a) ₹9,60,000
(b) ₹10,00,000
(c) ₹10,50,000
(d) ₹9,80,000

Answer: (b) Rationale: If Profit = 20% on Selling Price ($1/5$ of Sales), then Cost = 80% of Selling Price ($4/5$ of Sales). Profit on Cost = $1/4 = 25\%$. *Sales* = $₹8,00,000 \div 0.80 = ₹10,00,000$ (or ₹8,00,000 + 25% of ₹8,00,000 = ₹10,00,000).

SECTION 3: MATERIAL CONTROL, INVENTORY STOCK LEVELS & EOQ (Q16 – Q23)

Q16. At the Economic Order Quantity (EOQ) point, which two inventory costs are exactly equal to each other?

(a) Total Purchase Cost and Total Ordering Cost
(b) Total Ordering Cost and Total Carrying (Holding) Cost
(c) Total Carrying Cost and Stock-out Cost
(d) Prime Cost and Factory Overheads

Answer: (b) Rationale: The EOQ model minimizes total inventory holding expenditure precisely where annual ordering cost intersects and equals annual inventory carrying cost.

Q17. Annual demand = 10,000 units, Cost per order = ₹50, and Carrying cost per unit per annum = ₹4. The Economic Order Quantity (EOQ) is:

(a) 500 units
(b) 1,000 units
(c) 250 units
(d) 400 units

Answer: (a) Rationale: $EOQ = \sqrt{(2 \times A \times O) / C} = \sqrt{(2 \times 10,000 \times 50) / 4} = \sqrt{1,000,000 / 4} = \sqrt{250,000} = 500\text{ units}$.

Q18. Maximum consumption = 300 units/day, Minimum consumption = 100 units/day, Re-order period = 4 to 6 days. The Re-order Level (ROL) is:

(a) 1,200 units
(b) 1,800 units
(c) 600 units
(d) 1,500 units

Answer: (b) Rationale: $\text{Re-order Level (ROL)} = \text{Maximum Consumption} \times \text{Maximum Re-order Period} = 300\text{ units/day} \times 6\text{ days} = 1,800\text{ units}$.

Q19. The inventory control technique based on Pareto's Principle (80/20 rule), classifying inventory by monetary value of consumption, is:

(a) VED Analysis
(b) ABC Analysis
(c) FSN Analysis
(d) JIT System

Answer: (b) Rationale: ABC Analysis classifies inventory by value: 'A' items (high value, ~70% value, ~10% quantity), 'B' items (moderate value), and 'C' items (low value, ~10% value, ~70% quantity).

Q20. In an inflationary period characterized by rising raw material prices, which inventory valuation method results in lower reported profit and lower tax?

(a) FIFO (First-In, First-Out)
(b) LIFO (Last-In, First-Out)
(c) Simple Average Method
(d) Weighted Average Method

Answer: (b) Rationale: Under LIFO, the most recently purchased (higher priced) materials are issued to production, resulting in higher reported Cost of Goods Sold and reduced taxable profit.

Q21. The stock level below which inventory should never fall, and where emergency purchase procedures are immediately activated, is termed:

(a) Re-order Level
(b) Maximum Level
(c) Danger Level
(d) Buffer Stock

Answer: (c) Rationale: Danger Level = $\text{Average Daily Consumption} \times \text{Max Lead Time for Emergency Purchases}$. Stock reaching this trigger demands urgent replenishment.

Q22. In the VED inventory classification method commonly utilized for spare parts management, the acronym 'VED' stands for:

(a) Value, Economy, Delivery
(b) Vital, Essential, Desirable
(c) Variable, Estimated, Direct
(d) Verified, Evaluated, Documented

Answer: (b) Rationale: VED analysis categorizes engineering spares based on criticality to production: Vital (unavailability halts plant), Essential (serious loss), and Desirable (minor impact).

Q23. If Annual Demand is doubled while Ordering Cost and Carrying Cost per unit remain unchanged, the Economic Order Quantity (EOQ) will:

(a) Double
(b) Increase by 41.4% (i.e., by factor of \sqrt{2} \approx 1.414)
(c) Quadruple
(d) Remain unchanged

Answer: (b) Rationale: Because EOQ is proportional to $\sqrt{A}$, doubling demand yields $\sqrt{2} \times \text{EOQ} \approx 1.414 \times \text{EOQ}$, representing a 41.4% increase.

SECTION 4: SPECIFIC METHODS OF COSTING (JOB, PROCESS, CONTRACT & SERVICE) (Q24 – Q31)

Q24. Which method of costing is most appropriate for a commercial printing press, automobile repair workshop, and interior decorator?

(a) Process Costing
(b) Job Costing
(c) Operating (Service) Costing
(d) Continuous Output Costing

Answer: (b) Rationale: Job Costing is applied where production is non-standardized and executed strictly according to discrete customer specifications and individual job orders.

Q25. In pharmaceutical drug manufacturing and readymade garment production, where output is generated in discrete identifiable lots, the method used is:

(a) Batch Costing
(b) Single Output Costing
(c) Operating Costing
(d) Target Costing

Answer: (a) Rationale: Batch Costing is an extension of job costing where each cost unit is a batch/lot of identical products, optimized using Economic Batch Quantity (EBQ).

Q26. In Process Costing, unavoidable material loss inherent in the production nature (e.g., evaporation, shrinkage) is classified as:

(a) Abnormal Loss
(b) Normal Loss
(c) Abnormal Gain
(d) Capital Loss

Answer: (b) Rationale: Normal loss is an inevitable operational loss under normal manufacturing conditions. Its cost is absorbed by good units produced, inflating the per-unit cost of good output.

Q27. When the actual loss in a manufacturing process is less than the anticipated normal loss allowance, the resulting favorable variance is termed:

(a) Standard Gain
(b) Abnormal Gain / Abnormal Effectiveness
(c) Operating Profit
(d) Prime Gain

Answer: (b) Rationale: Abnormal gain occurs when actual output exceeds expected output (i.e., actual loss < normal loss). It is credited to the Abnormal Gain Account and transferred to Costing P&L.

Q28. In Contract Costing for construction projects, the portion of work certified by the architect/surveyor but retained by the contractee as a guarantee is:

(a) Progress Payment
(b) Retention Money
(c) Escrow Capital
(d) Work-in-Progress Reserve

Answer: (b) Rationale: Retention Money = $\text{Value of Work Certified} - \text{Payment Actually Received}$. It is retained until satisfactory defect-liability periods lapse.

Q29. A composite cost unit (combining two distinct dimensions such as quantity and distance) is utilized in which costing method?

(a) Job Costing
(b) Contract Costing
(c) Operating / Service Costing
(d) Process Costing

Answer: (c) Rationale: Operating costing utilizes composite units: Passenger-Kilometer (Passenger buses/Railways), Ton-Kilometer (Goods freight), Room-Day (Hotels), or Patient-Day (Hospitals).

Q30. A transport bus operates 5 trips daily between Srinagar and Baramulla (distance 50 km) carrying an average of 40 passengers. The daily Passenger-Kilometers total:

(a) 10,000 Passenger-km
(b) 20,000 Passenger-km
(c) 5,000 Passenger-km
(d) 8,000 Passenger-km

Answer: (a) Rationale: $\text{Daily Total Distance} = 5\text{ round trips} = 5 \times 50\text{ km} = 250\text{ km}$. $\text{Passenger-km} = 250\text{ km} \times 40\text{ passengers} = 10,000\text{ Passenger-km}$.

Q31. An escalation clause is routinely incorporated into long-term building construction contracts to protect the contractor against:

(a) Worker absenteeism
(b) Unanticipated price hikes in raw material and labor wage rates during contract tenure
(c) Architect inspection delays
(d) Liquidated damages from contractee

Answer: (b) Rationale: An escalation clause permits upwards adjustment of the contract price if market procurement rates for steel, cement, or statutory wages rise beyond agreed thresholds.

SECTION 5: MARGINAL COSTING & BREAK-EVEN / CVP ANALYSIS (Q32 – Q39)

Q32. Under Marginal Costing, the core equation governing Contribution is defined as:

(a) Sales - Total Cost
(b) Sales - Variable Cost (or Fixed Cost + Profit)
(c) Prime Cost + Factory Overhead
(d) Gross Profit - Administration Overhead

Answer: (b) Rationale: Contribution is the excess of sales revenue over variable cost ($S - VC$). It contributes first toward meeting fixed costs, and thereafter toward generating operating profit ($FC + Profit$).

Q33. If Selling Price = ₹100 per unit and Variable Cost = ₹60 per unit, the Profit-Volume (P/V) Ratio is:

(a) 60%
(b) 40%
(c) 25%
(d) 166.67%

Answer: (b) Rationale: $\text{Contribution} = ₹100 - ₹60 = ₹40$. $\text{P/V Ratio} = (\text{Contribution} / \text{Sales}) \times 100 = (40 / 100) \times 100 = 40\%$.

Q34. Fixed Cost = ₹2,00,000 and P/V Ratio = 40%. The Break-Even Point in terms of monetary sales value (Rupees) is:

(a) ₹5,00,000
(b) ₹80,000
(c) ₹8,00,000
(d) ₹2,80,000

Answer: (a) Rationale: $\text{Break-Even Sales (₹)} = \text{Fixed Cost} / \text{P/V Ratio} = ₹2,00,000 / 0.40 = ₹5,00,000$.

Q35. Total Actual Sales = ₹10,00,000 and Break-Even Sales = ₹6,00,000. What is the Margin of Safety (MoS) ratio?

(a) 60%
(b) 40%
(c) 25%
(d) 16.67%

Answer: (b) Rationale: $\text{Margin of Safety} = \text{Actual Sales} - \text{BEP Sales} = ₹10,00,000 - ₹6,00,000 = ₹4,00,000$. $\text{MoS Ratio} = (₹4,00,000 / ₹10,00,000) \times 100 = 40\%$.

Q36. Operating Profit earned by a business enterprise can be calculated directly by multiplying Margin of Safety (MoS) sales with:

(a) Gross Profit Ratio
(b) P/V Ratio
(c) Net Profit Ratio
(d) Fixed Cost Ratio

Answer: (b) Rationale: Because fixed costs are fully recovered at the Break-Even Point, all contribution generated beyond BEP converts directly into operating profit: $\text{Profit} = \text{Margin of Safety} \times \text{P/V Ratio}$.

Q37. Year 1: Sales = ₹4,00,000, Profit = ₹40,000; Year 2: Sales = ₹6,00,000, Profit = ₹80,000. The P/V Ratio is:

(a) 20%
(b) 25%
(c) 30%
(d) 40%

Answer: (a) Rationale: $\text{P/V Ratio} = (\text{Change in Profit} / \text{Change in Sales}) \times 100 = (₹40,000 / ₹2,00,000) \times 100 = 20\%$.

Q38. In a conventional Break-Even Chart, the point of intersection between the Total Revenue line and Total Cost line represents:

(a) Maximum Production Capacity
(b) Break-Even Point (BEP)
(c) Margin of Safety
(d) Angle of Incidence

Answer: (b) Rationale: The intersection point indicates Break-Even: Total Sales Revenue = Total Costs, indicating zero profit and zero operating loss.

Q39. On a Break-Even graph, the 'Angle of Incidence' is the angle formed between the:

(a) Total Sales Line and Total Cost Line at the Break-Even Point
(b) Fixed Cost Line and Horizontal Base Line
(c) Variable Cost Line and Vertical Axis
(d) Total Cost Line and Base Axis

Answer: (a) Rationale: The Angle of Incidence reflects profit-earning velocity once the break-even threshold is crossed; a wider angle indicates higher profitability per unit sold.

SECTION 6: STANDARD COSTING & VARIANCE ANALYSIS (Q40 – Q46)

Q40. When Actual Cost incurred is LESS than the predetermined Standard Cost, the variance is termed:

(a) Adverse (Unfavorable) Variance
(b) Favorable (Credit) Variance
(c) Null Variance
(d) Sunk Variance

Answer: (b) Rationale: An expenditure lower than standard or revenue higher than standard represents a Favorable (F) variance, indicating cost efficiency and higher profit.

Q41. Standard Price = ₹10/kg; Actual Price = ₹12/kg; Actual Quantity = 500 kg. What is the Material Price Variance (MPV)?

(a) ₹1,000 (Favorable)
(b) ₹1,000 (Adverse)
(c) ₹500 (Adverse)
(d) ₹1,200 (Favorable)

Answer: (b) Rationale: $\text{MPV} = \text{Actual Quantity} \times (\text{Standard Price} - \text{Actual Price}) = 500 \times (10 - 12) = -₹1,000 = ₹1,000\text{ Adverse (A)}$.

Q42. Standard Quantity = 400 kg; Actual Quantity used = 450 kg; Standard Price = ₹20/kg. What is the Material Usage (Quantity) Variance (MUV)?

(a) ₹1,000 (Adverse)
(b) ₹1,000 (Favorable)
(c) ₹900 (Adverse)
(d) ₹2,000 (Adverse)

Answer: (a) Rationale: $\text{MUV} = \text{Standard Price} \times (\text{Standard Quantity} - \text{Actual Quantity}) = ₹20 \times (400 - 450) = -₹1,000 = ₹1,000\text{ Adverse (A)}$.

Q43. The mathematical verification relationship among the primary material variances is expressed as:

(a) Material Cost Variance = Material Price Variance + Material Usage Variance
(b) Material Price Variance = Material Cost Variance + Material Usage Variance
(c) Material Usage Variance = Material Cost Variance × Material Price Variance
(d) Material Cost Variance = MPV - MUV

Answer: (a) Rationale: $\text{MCV} = \text{MPV} + \text{MUV}$. Material Cost Variance reconciles price fluctuations (purchasing efficiency) and consumption volume (shop-floor efficiency).

Q44. Which departmental manager is held primarily responsible for an unfavorable (adverse) Material Price Variance?

(a) Production Manager
(b) Purchase / Procurement Manager
(c) Sales Marketing Manager
(d) Chief Financial Officer

Answer: (b) Rationale: The Purchase Manager oversees vendor negotiations, order timing, and shipping terms, and is held accountable for deviations from standard purchase prices.

Q45. Standard wage rate = ₹50/hr, Actual wage rate paid = ₹55/hr, and Actual hours worked = 200 hours. The Labour Rate Variance (LRV) is:

(a) ₹1,000 (Favorable)
(b) ₹1,000 (Adverse)
(c) ₹2,500 (Adverse)
(d) ₹500 (Favorable)

Answer: (b) Rationale: $\text{LRV} = \text{Actual Hours} \times (\text{Standard Rate} - \text{Actual Rate}) = 200 \times (50 - 55) = -₹1,000 = ₹1,000\text{ Adverse (A)}$.

Q46. Idle Time Variance (due to machinery breakdown, power cuts, or raw material stock-outs) is ALWAYS:

(a) Favorable
(b) Adverse
(c) Zero
(d) None of the above

Answer: (b) Rationale: Idle time represents unproductive hours for which workers are compensated without generating output; hence, it is inherently an Adverse (cost loss) variance: $\text{Idle Hours} \times \text{Std Rate}$.

SECTION 6 (CONT.): MODERN COST CONCEPTS & CONTROL TECHNIQUES (Q47 – Q50)

Q47. The product pricing and costing philosophy that determines allowed production cost by deducting desired profit from competitive market price is:

(a) Cost-Plus Pricing
(b) Target Costing
(c) Historical Costing
(d) Marginal Mark-up Costing

Answer: (b) Rationale: Target Costing works backward: $\text{Target Cost} = \text{Anticipated Market Selling Price} - \text{Desired Profit Margin}$. Products are then engineered to meet this target.

Q48. The budgeting method that demands all operational activities be justified from scratch (zero base) for each new period, ignoring historical budgets, is:

(a) Incremental Budgeting
(b) Zero-Base Budgeting (ZBB)
(c) Flexible Budgeting
(d) Rolling Budgeting

Answer: (b) Rationale: Developed by Peter Pyhrr, Zero-Base Budgeting assumes a baseline of zero for every department; each expenditure must be freshly justified from the ground up.

Q49. A budget designed to change and dynamically adjust across multiple levels of operational activity is designated as a:

(a) Fixed Budget
(b) Flexible Budget
(c) Master Budget
(d) Performance Budget

Answer: (b) Rationale: A Flexible Budget recognizes the difference between fixed, semi-variable, and variable expenses, providing tailored cost benchmarks for different levels of capacity (e.g., 60%, 80%, 100%).

Q50. Under the Rowan Incentive Wage Plan, the bonus payable to a worker for saving time is computed using which formula?

(a) \text{Time Saved} \times \text{Standard Hourly Rate}
(b) 50\% \times (\text{Time Saved} \times \text{Hourly Rate})
(c) (\text{Time Saved} / \text{Time Allowed}) \times (\text{Time Taken} \times \text{Hourly Rate})
(d) \text{Time Taken} \times \text{Piece Rate}

Answer: (c) Rationale: Under Rowan: $\text{Bonus} = (\text{Time Saved} / \text{Standard Time}) \times (\text{Actual Hours Worked} \times \text{Rate})$. Halsey plan uses $50\% \times (\text{Time Saved} \times \text{Rate})$.

HIGH-YIELD EXAM REVISION SUMMARY: MASTER FORMULA CHECKLIST

Essential Mathematical Formulas for JKSSB / SSC CGL AAO:

- **EOQ:** $\sqrt{(2 \times A \times O) / C}$ (At EOQ: Annual Ordering Cost = Annual Carrying Cost).
- **Re-order Level:** $\text{Max Consumption} \times \text{Max Re-order Period}$.
- **Contribution:** $\text{Sales} - \text{Variable Cost} = \text{Fixed Cost} + \text{Profit}$.
- **P/V Ratio:** $(\text{Contribution} / \text{Sales}) \times 100 = (\text{Change in Profit} / \text{Change in Sales}) \times 100$.
- **Break-Even Point (BEP):** Units: $\text{Fixed Cost} / \text{Contribution per unit}$; Value (₹): $\text{Fixed Cost} / \text{P/V Ratio}$.
- **Margin of Safety:** $\text{Actual Sales} - \text{BEP Sales} = \text{Profit} / \text{P/V Ratio}$.
- **Material Variances:** $\text{MCV} = (\text{SQ} \times \text{SP}) - (\text{AQ} \times \text{AP})$; $\text{MPV} = \text{AQ} \times (\text{SP} - \text{AP})$; $\text{MUV} = \text{SP} \times (\text{SQ} - \text{AQ})$.